GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Comprehensive GATE Notes: Control Systems

Paper Code: EC • Section 6: Control Systems (8-10 Marks)

Subject Code: EC-CTRL • **Target Exam:** GATE ECE • **Weightage:** 8-10 Marks

1. Second-Order Time Domain Response & Damping

- **Standard System:** T(s) = omega_n^2 / (s^2 + 2*zeta*omega_n*s + omega_n^2).
- **Key Parameters (0 < zeta < 1):**
• Peak Time: t_p = pi / (omega_n sqrt(1 - zeta^2)).
• Peak Overshoot: %M_p = 100 * e^{-pi * zeta / sqrt(1 - zeta^2)}.
• 2% Settling Time: t_s = 4 / (zeta * omega_n).

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Peak Time t_p	**t_p = pi / (omega_n sqrt(1 - zeta^2))**	omega_n = Natural Freq, zeta = Damping	Time to reach the very first overshoot peak
Max Overshoot %M_p	**%M_p = 100 * exp(- pi * zeta / sqrt(1 - zeta^2))**	zeta = Damping Ratio	Function solely of damping ratio zeta
2% Settling Time	**t_s = 4 / (zeta * omega_n)**	zeta * omega_n = Real Part of Pole	Four time-constants of dominant envelope

GATE PYQ (2 Marks) - Damping & Overshoot Calculation
Question: Unity feedback system has G(s) = 25 / [s(s+6)]. Find omega_n, zeta, and %M_p.

Step 1: T(s) = 25 / [s^2 + 6s + 25].
Step 2: omega_n^2 = 25 => **omega_n = 5 rad/s**. 2*zeta*5 = 6 => **zeta = 0.6**.
Step 3: %M_p = e^{- (pi * 0.6) / 0.8} * 100% = e^{-2.356} * 100% = **9.48%** ≈ **9.5%**.

Essential GATE Exam Pitfalls & Scoring Traps

- **Important:** Root locus branches always start at open-loop poles and terminate at open-loop zeros.
- **Important:** Both GM > 0 dB and PM > 0 deg must hold for closed loop stability.